In [2]:
```
orders = [
{
'id': 101,
'customer': 'Asha',
'status': 'DELIVERED',
'items': [('keyboard', 2, 1499.0), ('mouse', 1, 599.0)]
},
{
'id': 102,
'customer': 'Ravi',
'status': 'CANCELLED',
'items': [('monitor', 1, 10999.0)]
},
{
'id': 103,
'customer': 'Asha',
'status': 'DELIVERED',
'items': [('cable', 5, 199.0)]
},
{
'id': 104,
'customer': 'Meera',
'status': 'SHIPPED',
'items': [('keyboard', 1, 1499.0), ('monitor', 1, 10999.0)]
},
{
'id': 105,
'customer': 'Ravi',
'status': 'DELIVERED',
'items': [('mouse', 3, 599.0)]
}
]

# A. Customer Revenue

def analyse(orders):
    revenue = {}

    for order in orders:

        if order['status'] != 'CANCELLED':

            customer = order['customer']

            if customer not in revenue:
                revenue[customer] = 0

            for product, quantity, price in order['items']:
                revenue[customer] += quantity * price

    print("Revenue:")
    print(revenue)
```

```
    # B. Top Selling Products
    products = {}

    for order in orders:

        if order['status'] != 'CANCELLED':

            for prod, quan, price in order['items']:

                if prod not in products:
                    products[prod] = 0

                products[prod] += quan

    top_products = sorted(
        products.items(),
        key=lambda x: (-x[1], x[0])
    )[:2]

    print("\nTop Products:")
    print(top_products)


    # C. Fully Delivered Customers
    customer_status = {}

    for order in orders:

        customer = order['customer']

        if customer not in customer_status:
            customer_status[customer] = []

        customer_status[customer].append(order['status'])

    fully_delivered = sorted(
        customer
        for customer in customer_status
        if all(status == 'DELIVERED'
               for status in customer_status[customer])
    )

    print("\nFully Delivered Customers:")
    print(fully_delivered)

    result = {
        'revenue': revenue,
        'top_products': top_products,
        'fully_delivered': fully_delivered
    }
print("\nFinal Output:")
result = analyse(orders)
print(result)
```

```
Final Output:
Revenue:
{'Asha': 4592.0, 'Meera': 12498.0, 'Ravi': 1797.0}

Top Products:
[('cable', 5), ('mouse', 4)]

Fully Delivered Customers:
['Asha']
None
```

In [1]:

```python
lines = [
    ' asha VERMA ; ASHA.V@Example.COM ; +91 98765-43210 ;pune ',
    'ravi kumar;ravi@@corp.in;9123456780;Mumbai',
    'meera;meera@mail;9123456780;Delhi',
    'john doe; john@x.org ;12345; Goa',
    'Neha Rao;neha.rao@firm.co.in;919876501234;nashik',
    'amit;amit@site.com;9876543210'
]
def clean_customer(raw):
    data = raw.split(';')

    # Check exactly 4 fields
    if len(data) != 4:
        return None

    name = data[0].strip()
    email = data[1].strip().lower()
    phone = data[2].strip()
    city = data[3].strip()

    # Validate name
    if name == '':
        return None

    words = name.split()

    for word in words:
        if not word.isalpha():
            return None

    name = name.title()

    # Validate city
    if city == '' or not city.replace(' ', '').isalpha():
        return None

    city = city.title()

    # Validate email
    if email.count('@') != 1:
        return None

    parts = email.split('@')
    username = parts[0]
    domain = parts[1]
```

```
    if username == '':
        return None

    if '.' not in domain:
        return None

    if domain.startswith('.') or domain.endswith('.'):
        return None

    # Clean phone number
    phone = phone.replace(' ', '')
    phone = phone.replace('-', '')

    if phone.startswith('+91'):
        phone = phone[3:]
    elif len(phone) == 12 and phone.startswith('91'):
        phone = phone[2:]

    # Validate phone
    if len(phone) != 10:
        return None

    if not phone.isdigit():
        return None

    if phone[0] not in '6789':
        return None

    # Return dictionary
    return {
        'name': name,
        'email': email,
        'phone': phone,
        'city': city
    }

rejected = 0

for line in lines:
    result = clean_customer(line)

    if result is not None:
        print(result)
    else:
        rejected = rejected + 1

print("Rejected:", rejected)
```

```
{'name': 'Asha Verma', 'email': 'asha.v@example.com', 'phone': '9876543210', 'city':
'Pune'}
{'name': 'Neha Rao', 'email': 'neha.rao@firm.co.in', 'phone': '9876501234', 'city':
'Nashik'}
Rejected: 4
```